Tier 2 | Approximation and compression

Choose the search shortcut

LSH groups likely neighbors by hashes. IVF searches selected centroid partitions. HNSW navigates a neighbor graph. Each saves work by risking missed neighbors; exact search remains the reference.

Tune one control at a time

LSH: tables/bits/probes. IVF: nlist/nprobe. HNSW: M/efConstruction/efSearch. Start with query effort, then rebuild if no quality/cost point qualifies. Do not tune on final evaluation queries.

Compress with explicit costs

Scalar quantization reduces coordinate precision. PQ learns subvector codebooks. Binary methods store bits. Count codebooks, graph links, IDs and retained full vectors; a one-bit code is not the entire service footprint.

Filters and hybrid retrieval

Compute truth inside the eligible set. Postfiltering a small global candidate list can miss nearly every eligible neighbor. Fuse dense and lexical ranks with RRF when raw scores are incompatible: contribution = 1/(constant + rank).

Recorded caution

The first scratch HNSW fixture scored 0.896 against a 0.900 target. A higher-budget validation reached 1.000 with greater memory and latency. Read the workload details in docs/PHASE_4_REPORT.md before generalizing.

Run and explain

uv run --extra libs python scripts/run_bench.py configs/bench/faiss_exact.yaml
Use the existing configs/bench files for your installed backend. Report recall, p95, errors, build time, resident memory and persisted bytes. See docs/INDEX_TUNING_GUIDE.md.